Q1 :- a) unexpected finding

The unusual result is premium customers with high satisfaction still have high churn

Value = 70

It's unexpected because Normally, when Satisfaction is high, churn should be low.

Possible Reasons :-

* Premium customers may get better offers from competitors.
* Premium accounts may have high fees.
* Customers may ~~leve~~ leave for Reasons other Than Satisfaction, like Location, Pricing, digital, services or policy changes

b) Highest churn Rate / highest churn Value

⇒ Highest churn = 100

Category factors = savings account, Low satisfaction.

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Q2 :- Score 3 → 10
Score 4 → 132
Score 5 → 53

$N = 10 + 132 + 53 = 195$

a) Mean $\bar{x} = \frac{\sum f(x)}{\sum f}$

$$\bar{x} = \frac{(3)(10) + (4)(132) + (5)(53)}{195} = 4,22$$

$$SD = \sqrt{\frac{\sum f (x - \bar{x})^2}{N}} = 0,52$$

b) Skewness

Most of the scores are high : 4 and 5 and only small Numbers at the Lower side : 3

So The data is piled up on The right with a tail on The left

So → The distribution is slightly left [left /Negative skewness]

Comments → • Most customers gave high Ratings
• only a few gave Lower Ratings.

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c) Hurghada → Mean = 4.22 , SD = 0.52 $\boxed{3}$

Sharm → Mean = 4.2 , SD = 0.05

Taba → Mean = 4.45 , SD = 0.60

→ Hurghada VS Sharm:

Means are almost The same but sharm has a Much smaller SD. both hotel shave Similar avg Ratings but sharm is Much Consistent.

→ Hurghada VS Taba:

Taba has a higher mean , Also Taba has Slighty Lower spread. This Means Taba Preform better overall and also slighty Consistent than hurghada.

Q3:-

Note That * Correlation close to +1 = strong Positive Relationship

* higher positive correlation = stronger association

* But correlation close Not prove Causation

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a) Store Availability and Customer Reviews.

→ Store availability is More Stronger associated with sales volume Than customer Reviews because 0,80 > 0,65. This Means store availability is More imperlent in explaning Sales.

b) Social Media Engagement and Sales Volume

Correlation = 0,75

→ Strong positive Correlation. But we cannot Conclude direct causation.

Because correlation only Tells us That two Variables move Together. It dose Not prove That one causes The other.

→ Also another variable May affect both such as advertising budget.

So A Correation of 0,75 only shows a Strong positive relationship Not Cause and effect.

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c) Effect of outliers :-

Outliers can distort The correlation coefficient by Making The Relationship appear stronger or weaker Than it actually is. Theirimpact can be Reduced by checking and removing erroneous outliers.

d) Fan shap ,

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